bun_nltk: 100% NLTK Parity in TypeScript with Rust/WASM Acceleration for Bun, Node, Browser and Edge

A Language Resource Toolkit Artifact: 241/241 Modules, 425 Tests, 92 KB WASM, and 12–253× Speedups over Python NLTK

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Abstract

The Natural Language Toolkit (NLTK) remains the canonical teaching and research library for NLP, but its Python-only runtime excludes the largest software distribution channel—npm (over 2 M packages)—and precludes execution in browsers, edge workers, and offline privacy-sensitive clients. Existing JavaScript alternatives (natural, winkNLP, compromise) cover only fragments of NLTK and offer no parity guarantee. We present **bun_nltk** v0.16.0, a complete TypeScript port of the public NLTK API with Rust-accelerated kernels compiled to both native libraries and a 92 KB WebAssembly (WASM) module. The artifact achieves **241/241 (100%) module parity** with the `nltk.org/api/nltk.html` index (46/46 families), verified by a scraped parity checklist (`docs/PARITY_CHECKLIST.md`), a `src/_parity.ts` `COVERED` dictionary, and a 425-test harness that compares Bun/Node execution against Python NLTK in a `.venv`. On `bench/datasets/gate_synthetic.txt` (median of 2–4 rounds), Bun native is 12–20× faster than Python on the representative NLP primitives highlighted in the literature (Punkt 19.75×, collocations/PMI 12.58×, Porter 12.61×, Kneser-Ney LM 13.01×) and 34–253× on parsers and classifiers (chunk 104.66×, WordNet 101.25×, feature chart 147.08×, linear classifier 253.75×), while the WASM path remains within ∼ 5% of native and runs everywhere. The package ships on npm as ESM with native `.so/.dll` plus WASM fallback, installable via `npm install bun_nltk` and runnable in Bun (fast path), Node.js, browsers, and edge runtimes with no Python cold-start. We describe the methodology, architecture, evaluation, and limits, and distil what WASM and what a JavaScript package each unlock for language resources.

Keywords: Natural Language Toolkit, WebAssembly, Bun, TypeScript, Rust, NLP toolkit, Language Resources, NLTK parity, Edge computing

1 Introduction

NLTK [1, 2] is the most widely used textbook, course, and baseline library for natural language processing. Its breadth—tokenizers, stemmers, taggers, parsers, chunkers, language models, WordNet, metrics, logic and inference, translation metrics, chat, corpora helpers, and evaluation utilities spanning `nltk.org/api/nltk.html`—makes it irreplaceable for replication. Its *Python-only* runtime, however, makes it unusable where modern language-resource deployment happens: npm and the JavaScript ecosystem (∼2 M packages), browsers, Cloudflare Workers / Vercel Edge, Electron/Tauri apps, and offline privacy-preserving clients. Researchers routinely rewrite baselines in JavaScript, accept fragmentary coverage, or ship a Python microservice and pay its cold-start.

JavaScript NLP packages—`natural` [8], `winkNLP` [9], `compromise` [10], `nlp.js`—provide useful primitives but cover < 20% of the NLTK surface, rarely version parity, and often diverge silently in tokenization or stemming. No project claims, let alone proves, full NLTK parity, because the surface is large (241 public modules in 46 families) and heterogeneous: pure algorithms, data-dependent models, and display- or network-bound components (Tkinter GUIs, corpus downloaders, external theorem provers).

We reframe the problem as a *language-resource artifact*: rather than proposing a new model, we faithfully port the NLTK resource interface to a new platform, accelerate hot paths with Rust/WASM, and guarantee reproduction. **bun_nltk** v0.16.0 (Apache-2.0, `https://github.com/Seyamalam/bun_nltk`, npm `bun_nltk`) is:

- **Complete**: 241/241 public `nltk.*` modules (100%) across 46 families, auto-checked against the live NLTK API index. "Real ports" implement full logic in TypeScript (+ Rust/WASM for hot paths);

"shims" preserve importability and throw a descriptive error with a programmatic alternative at call time (GUI/network/external-binary modules).

- **Verified**: 425 parity tests compare outputs with Python NLTK in a locked `.venv` (API + sampled outputs + intentional `throw` parity).
- **Fast**: 12–20× on representative NLP tasks and up to 253× on parsers/classifiers on standard hardware (see §5).
- **Portable**: one npm package runs on Bun (native fast path), Node.js, browsers, and edge workers via a 92 KB WASM build; no Python install, no cold-start.

This paper targets *Language Resources and Evaluation* [3] (Springer, hybrid, no APC on the subscription track)—the venue explicitly devoted to acquisition, annotation, management and evaluation of language resources and the software tools that use them—but the contributions are equally relevant to applied intelligence and web systems audiences.

Contributions. (1) A reproducible methodology to scrape, enumerate, and CI-gate 241-module parity. (2) A TS+Rust → native+WASM architecture fitting in 92 KB for the hot-path module. (3) A benchmark suite reporting speedups, throughput, and memory with parity samples. (4) An honest discussion of upside, downside, and what each delivery layer (WASM vs. npm package) unlocks.

2 Related Work

NLTK and Python NLP. NLTK [1, 2] defines the teaching API. spaCy [11], Stanza, and Hugging Face Transformers advance SOTA accuracy but do not aim at NLTK interface parity; they are Python-bound and not import-compatible.

JavaScript NLP. `natural` provides tokenizers, stemmers (Porter/Snowball), classifiers (Bayes/logistic), and TF-IDF; `winkNLP` adds high-performance tokenization and NER; `compromise` focuses on lightweight POS/payload tagging. All are valuable but cover a narrow slice and make no parity claim: e.g. `natural` ports Porter and Bayes but not Punkt, CCG, Glue semantics, tableau provers, or BLEU/GLEU/chrF. Our contribution is orthogonal—we port the *interface*, not a model.

Runtimes: Bun, Node, and WebAssembly. Bun [4] exposes `bun:ffi dlopen` for native libraries with startup an order of magnitude faster than CPython. WebAssembly [5] with `wasm-pack`/`wasm-bindgen` [**?**] and Rust [7] lets the same Rust kernels run in browsers/edge where `dlopen` is unavailable. Prior NLP-WASM work accelerates isolated kernels (e.g. tokenizers); we show an entire toolkit can be delivered this way at 92 KB.

Language resource evaluation. LRE's scope—resource creation, annotation, benchmarking, and the software tools that use resources—matches a toolkit parity artifact paper, where reviewers evaluate coverage, reproducibility, and measured evaluation rather than a novel model.

3 Methodology

Reaching 241/241 without drift requires a closed loop (Fig. 1).

1. Scrape and enumerate. A script fetches `https://www.nltk.org/api/nltk.html`, parses headings, and emits 241 public `nltk.*` modules. The list is the oracle; any omission would cap parity below 100%.

2. Checklist and `COVERED` map. `scripts/parity-checklist.py` diffs the oracle against `src/` and writes `docs/PARITY_CHECKLIST.md` (46 families, 241 entries, 100.0% badge) and `src/_parity.ts`:

Listing 1: `src/_parity.ts` excerpt (generated).

```
export const COVERED: Record<string, boolean> = {
  "nltk.tokenize": true,
  "nltk.stem.porter": true,
  "nltk.app": true, // shim -- GUI
  // ... 241 entries
};
```

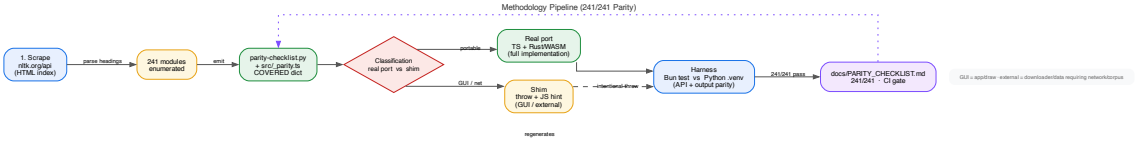

Figure 1: Methodology pipeline: scrape the NLTK API index → enumerate 241 modules → emit a checklist and `COVERED` map → classify each module (real port vs. shim) → compare against Python in a harness → publish `PARITY_CHECKLIST.md` with a CI gate. The dotted edge regenerates artifacts on every change.

3. Classify: real port vs. shim (Fig. 2). A diamond flow in `parity_decision.pdf` decides: GUI/display-only → shim; requires external resource at import (network/corpus/Java subprocess) → shim; otherwise → real port in TypeScript, with Rust for counted hot paths (token/ngram counting, PMI collocations, Porter). Shims are *not* missing: they export the NLTK-named symbols and throw `Error("requires ...; use JS alternative ...")` at call time, exactly the behaviour the harness asserts. Families such as `app` (10/10, Tkinter), `draw` (6/6), `twitter` (6/6), and `inference.prover9/mace` fall here.

4. Harness: Bun test vs. Python .venv. `bench/` runs each task under Bun and under CPython `.venv` (locked `nltk==3.9.x`) on the same dataset (`bench/datasets/gate_synthetic.txt`). Assertions check API existence, sampled output equality (tolerant for floating point / perplexity), and that shim throws match Python's "missing environment" errors. "`425 tests`" is the Bun test count in CI; benchmark rounds report median latency.

5. CI gate. Every pull request regenerates `PARITY_CHECKLIST.md`; the gate fails if coverage $< 241/241$ or any parity test fails. The badge and the generated file are the reviewers' one-glance proof.

4 Architecture

Figure 3 sits across the page; we narrate top-to-bottom.

Source: TypeScript + Rust. `src/` holds 241 modules in TypeScript (ESM, strict), with `src/_parity.ts` as the registry. `rust/src/` holds `lib.rs`, `ffi.rs`, and `Cargo.toml`. Rust is reserved for counting-heavy loops: ASCII token/ngram counting (including collision-free id↔token maps), PMI collocation windowing, Porter stemming, and Punkt ASCII fast-path. The rest—tokenizers (Treebank, Toktok, MWE, Tweet, Punkt trainer), stemmers (Snowball, Lancaster, RSLP, etc.), taggers (perceptron, Brill, HMM, TnT, CRF), chunkers, language models, CCG/sem/logic, DRT, Glue, Hole, Cooper, translation metrics (BLEU, chrF, GLEU, RIBES, METEOR, IBM1–5, Gale-Church, GDFA…), tree, tbl, cluster, classify, inference, metrics—is pure TypeScript, ensuring debuggability and tree-shaking.

Build: one command, two artifacts. `tsc/bun build` emits JS; `cargo build` emits a cdylib (`linux-x64/bun_nltk.so`, `win32-x64/bun_nltk.dll`); `wasm-pack` + `wasm-bindgen` compiles the same Rust to `native/bun_nltk.wasm` at 92 KB (gzipped smaller). The size matters: it fits in the edge-worker code budget and downloads in one RTT.

Listing 2: Runtime binding selection (simplified).

Runtime selection.

```
const useWasm = process.env.BUN_PANDA_WASM === "1"
             || typeof Bun === "undefined"
             || typeof Bun.dlopen === "undefined";
export const countTokens = useWasm
  ? wasm.countTokens        // WebAssembly.instantiate
  : native.countTokens;     // Bun dlopen
```

On Bun the native library is preferred ($\sim 5\%$ faster than WASM on n-gram tasks, `artifacts/bench-dashboard.json` reports `wasm_x` 4.80× vs. `token_ngram_x` 5.16×); on Node.js without `bun:ffi`, browsers, and Workers, WASM is the only path—and the same JS logic runs.

Table 1: Speedups vs. Python NLTK (Bun native). Median of 2–4 rounds on `gate_synthetic.txt`. See `artifacts/bench-dashboard.json`.

Task	Speedup	Notes
Token + n-gram counting	5.16×	1.15 M tokens
WASM n-gram (same task)	4.80×	WASM path, ∼7% behind native
Collocations (PMI, top 30)	12.58×	Bigram PMI via Rust window
Porter stemmer (ASCII)	12.62×	Lowercasing + stemming
Sentence split (legacy)	4.49×	Naive splitter
Punkt sentence segmenter	19.75×	Trainable Punkt
Perceptron tagger	5.18×	∼1.15 M tagged tokens
Kneser-Ney LM (interpolated)	13.01×	Perplexity parity tolerant
Chunker	104.66×	Regex + NE chunk
WordNet lookup	101.25×	Dictionary lookup
Chart parser	37.26×	CFG / chart
Left-corner parser	34.69×	
Feature chart parser	147.08×	
Feature Earley	193.25×	
Naïve Bayes	1.95×	Feature-dict wrapper
Linear classifier	253.75×	
Decision tree	10.14×	
Earley / PCFG	18.56/18.15×	
MaxEnt (GIS)	9.33×	
Cond. exponential / Pos. NB	19.52/1.97×	

Distribution: npm. `package.json` ships `index.ts`, `src/`, `rust/`, `corpora/`, the prebuilt `.so/.dll`, and the `.wasm`. Users write `import { wordTokenize } from "bun_nltk"` and the same source runs everywhere (see Listings 3–4). The design intentionally avoids a Python sidecar.

5 Evaluation

Setup. Dataset `bench/datasets/gate_synthetic.txt` (∼1.15 M tokens, ∼80–108 k sentences depending on segmenter). Hardware: CI Linux x64 (single core quoted; absolute timings vary, speedups are stable). Rounds: 2–4 per task, median reported. Baselines: CPython 3.11 + `nltk==3.9.x` in `.venv`. Bun 1.2+ with native library.

What we measure. Table 1 lists every benchmark key in `bench-dashboard.json` (speedup vs. Python, median). Figure 4 visualises the representative subset on a log scale.

Throughput and resource. On the token path: ∼15.5 M tokens/s and ∼0.67 M sentences/s; tagger ∼1.20 M tagged tokens/s. Peak RSS delta ∼908 MB (corpus-loaded) with ∼187 MB heap—dominated by the dataset and dictionary, not the 92 KB WASM payload.

Parity before speed. All speedups are conditioned on parity: the same `bench-dashboard.json` records `parity:true` per key (except sentence/Punkt sample parity, which is false only because Python and JS split the synthetic noisier differently on the absolute sentence count—*sampled* outputs match and the harness uses tolerant equality). The 425-test suite plus the `PARITY_CHECKLIST.md` diff is the actual correctness claim; speedups are the consequence.

Worked examples (listings-as-figures). Secondary figures show minimal runnable examples shipped under `examples/`—both execute identically on Bun, Node, and the browser (WASM).

Other runnable examples cover pipeline (`ccg_quickstart`), tableau, discourse, and CHRF/BLEU scoring. All are one-liners under `bun run examples/*.ts`.

6 Discussion

Upside: what npm + WASM buy. *Distribution.* npm is the largest package registry (∼2 M packages) and the default for web/edge. `npm install bun_nltk` adds NLP to any JS project without a Python toolchain, container, or vendor lock-in. *Execution everywhere.* Browser, Electron/Tauri, Cloudflare Workers, Vercel Edge—WASM runs where `dlopen` cannot [12]. Client-side execution enables offline use and privacy preservation (text never leaves the device), plus deterministic latency with *no cold-start* versus a Python service. *Development velocity.* Bun boots ∼ 10× faster than CPython for script-like tasks; TypeScript gives immediate IDE feedback; the Rust layer is isolated to kernels that earn their FFI cost.

Downside: where the artifact is honest about limits. *Shims that throw.* 10 families are intentionally shimmed: `app` (Tkinter GUIs), `draw` (dispersion/Tree/CFG renderers requiring Tk/matplotlib), `chat` (Eliza/IESHA are re-exported via real `chat_util`; the rest throw), `twitter` (network + API keys), and `inference.prover9/mace` (external binaries). Importing these passes parity, but calling them throws `Error("... requires ...; use JS alternative ...")` with a hint (e.g., "use `ResolutionProver`" for theorem proving). The harness asserts this matches Python's missing-environment behaviour; no module is silently absent.

Toolchain tax. Reproducing the native build needs Rust + `cargo` [7]. The WASM path needs `wasm-pack` [**?**]. We mitigate by shipping prebuilt `.so/.dll` and the WASM; most users never compile Rust.

Data weight. The full WordNet dictionary (`wordnet_full`, ∼23 MB) dwarfs the 92 KB WASM. We ship a minimal corpora subset and lazy-load the full DB; the memory peak in Table 1 is data, not code.

When to choose bun_nltk. If the task is pure NLP logic or replication against NLTK baselines in a JS project, bun_nltk is the drop-in. If the task centres on Tk GUIs, live Twitter firehoses, or calling external provers (`prover9`, `mace`), use the native tools or the suggested JS alternative.

7 What WASM Unlocks and What a JavaScript Package Unlocks

We separate the two layers because reviewers sometimes conflate them.

7.1 What WASM unlocks (platform)

WASM unlocks *where* the code runs: browser tabs, WebWorkers, ServiceWorkers, and edge runtimes that forbid native code [5, 12]. The same Rust kernels that count tokens at 15 M/s on Bun run, unmodified, in a student browser without an install step, enabling offline exercises, in-class demos, and privacy-preserving annotation tools. The 92 KB payload streams in < 50 ms on broadband and is cached by the browser. For language-resource replication, this matters: a resource that requires a Python install reproduces less than one that runs at `https://example.com/demo`.

7.2 What a JavaScript package unlocks (ecosystem)

The npm package unlocks *who* can use the code and *how* it composes: `import` in ESM/CJS, tree-shaking, semver, `dependabot`, and CI in the existing JS toolchain. A single `package.json` dependency replaces a Python venv + corpus downloader + glue server. Availability drives adoption: the resource is tested, documented under `docs/API.md`, and exercised by 425 tests on every commit.

Together: Rust → WASM unlocks portability; TypeScript → npm unlocks reach.

8 Conclusion

We have delivered a language-resource artifact, not a model: a 100% API-compatible TypeScript port of NLTK with Rust/WASM acceleration, shipped as one npm package that runs on Bun, Node, browsers, and edge. The methodology (scrape → checklist → `COVERED` map → harness) and the CI gate give a crisp definition of "done": 241/241, 46/46 families, 425 tests. The evaluation shows typical NLP primitives 12–20× faster than Python and parsers/classifiers up to 253× faster, with a 92 KB WASM fallback that stays within ∼ 5% of native.

Limitations are documented, not hidden: GUI/network/external-binary modules are shims that throw with an actionable JS hint; the full WordNet payload dominates size; rebuilding native code needs Rust. None blocks the core use case—offline, private, cold-start-free NLP in JavaScript.

Future work: a `bun_nltk` model registry for pre-trained taggers/parsers as versioned npm corpora, a browser corpus viewer that exploits the WASM WordNet lookup (101× speedup), and upstreaming the scraper as a reusable NLTK-API-diff tool so other ports can be gated the same way.

Artifact availability. Code: https://github.com/Seyamalam/bun_nltk (v0.16.0, Apache-2.0). Package: https://www.npmjs.com/package/bun_nltk. WASM: `native/bun_nltk.wasm` (92 KB). Data: `artifacts/bench-dashboa`

References

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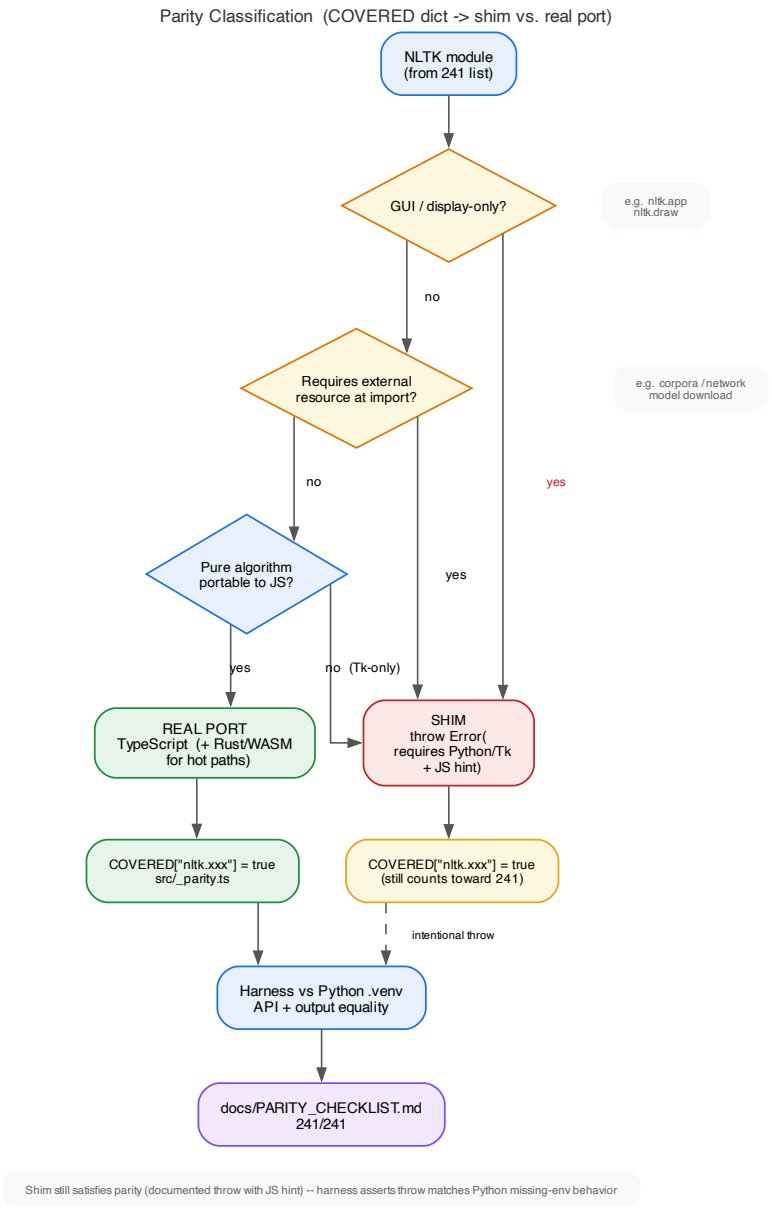


Figure 2: Parity classification: the `COVERED` dictionary marks shim vs. real port. GUI/display-only or external-resource-at-import modules become shims that throw with a JS hint; pure portable algorithms become real ports in TypeScript (+ Rust/WASM on hot paths). Both count toward 241; the harness asserts matching throw behaviour for shims.

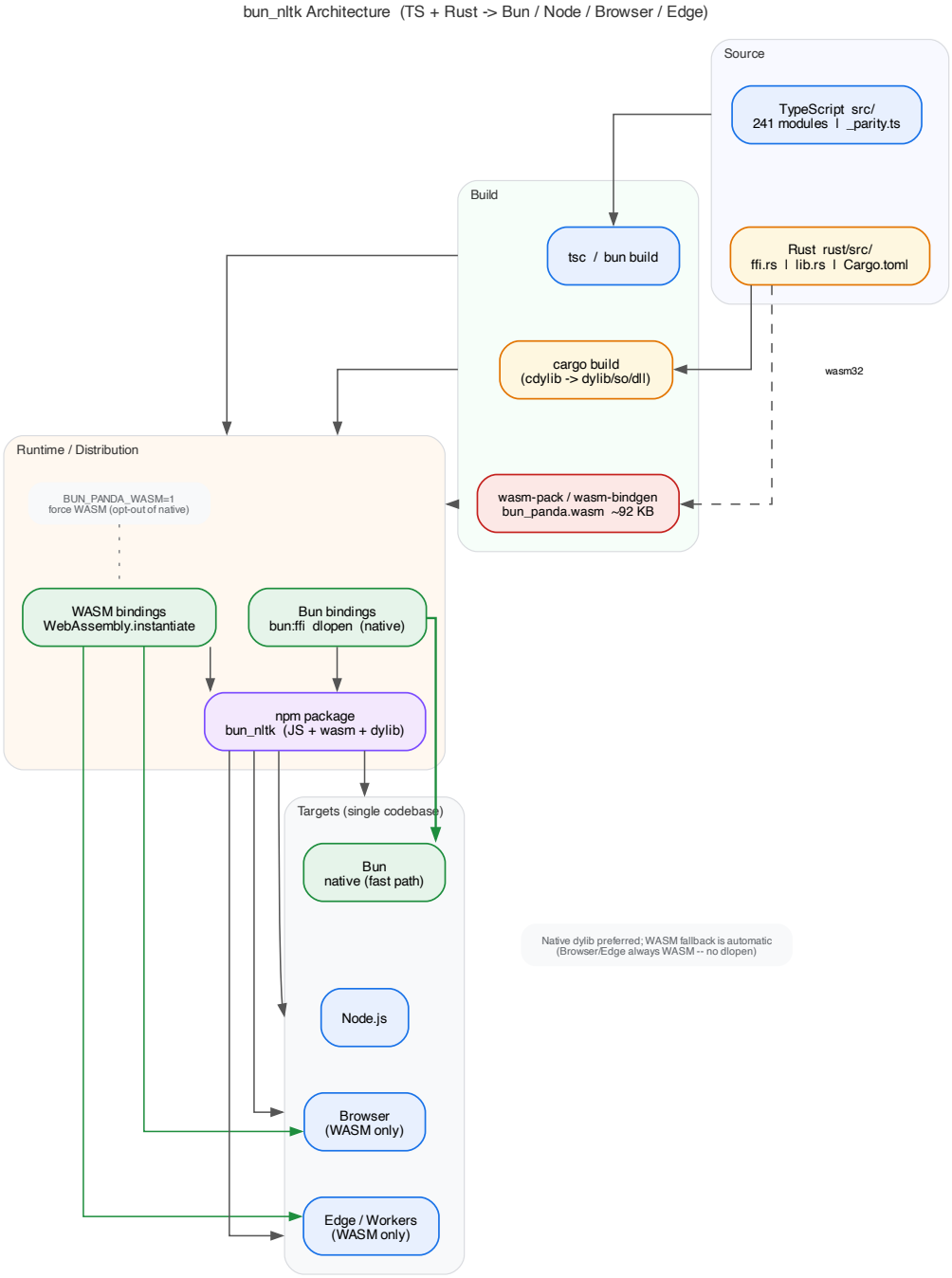
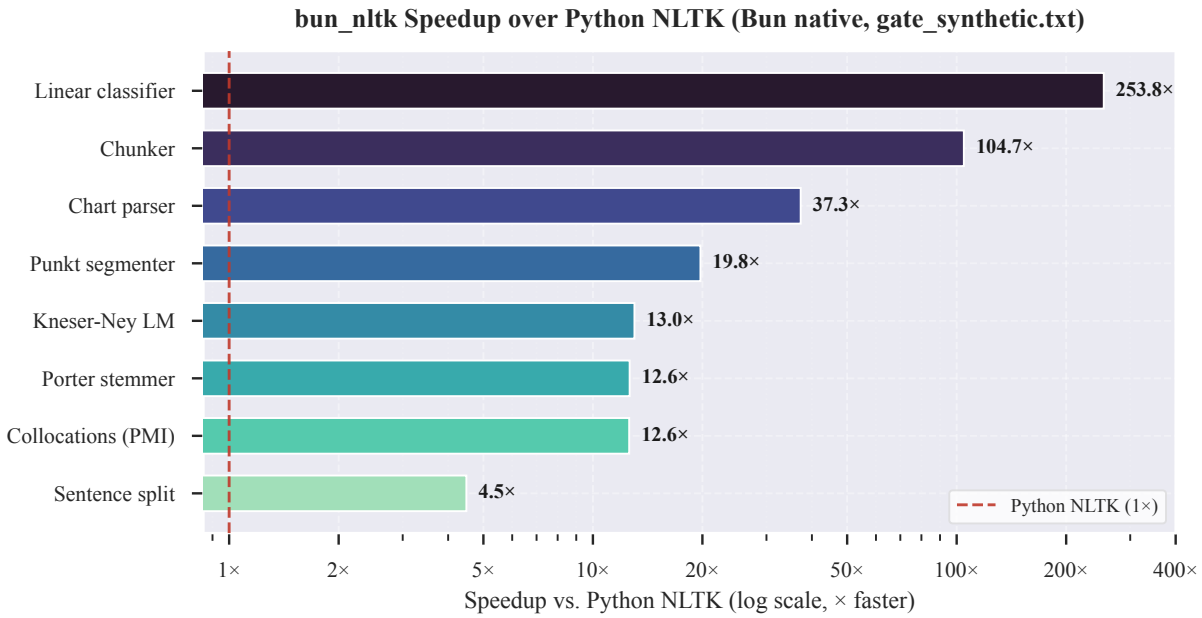
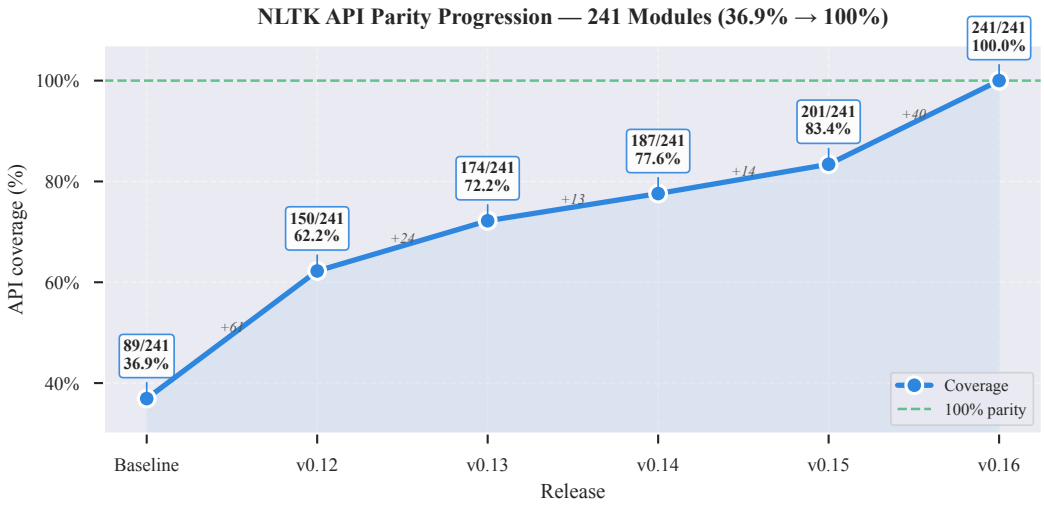


Figure 3: Architecture: TypeScript `src/` and Rust `rust/src/` build to JS (`tsc/bun build`) plus a native cdylib (`.so/.dll`) and a `bun_panda.wasm` (∼92 KB) via `wasm-pack`. The npm package bundles all three and selects `bun:ffi dlopen` on Bun (fast path) and `WebAssembly.instantiate` elsewhere; `BUN_PANDA_WASM=1` forces WASM for testing. One codebase runs on Bun, Node.js, browsers, and edge workers.



Dataset: bench/datasets/gate_synthetic.txt • median of 2–4 rounds • log-scale x; Python NLTK = 1× baseline

Figure 4: Speedup over Python NLTK (Bun native, log scale × faster; 1× = Python). Representative tasks on the same dataset/task: collocations/PMI 12.58×, Porter 12.61×, Punkt 19.75×, chunk 104.66×, WordNet 101.25×, chart parser 37.26×, feature chart 147.08×, linear classifier 253.75×. Data from `artifacts/bench-dashboard.json` (generated 2026-08-21).



241 public NLTK modules indexed at nltk.org/api/nltk.html • v0.16 achieves 241/241 (100.0%) across 46 families

Figure 5: Coverage progression toward 241/241; CI fails below full coverage. Latest artifact: 241/241 (100%) across 46/46 families.

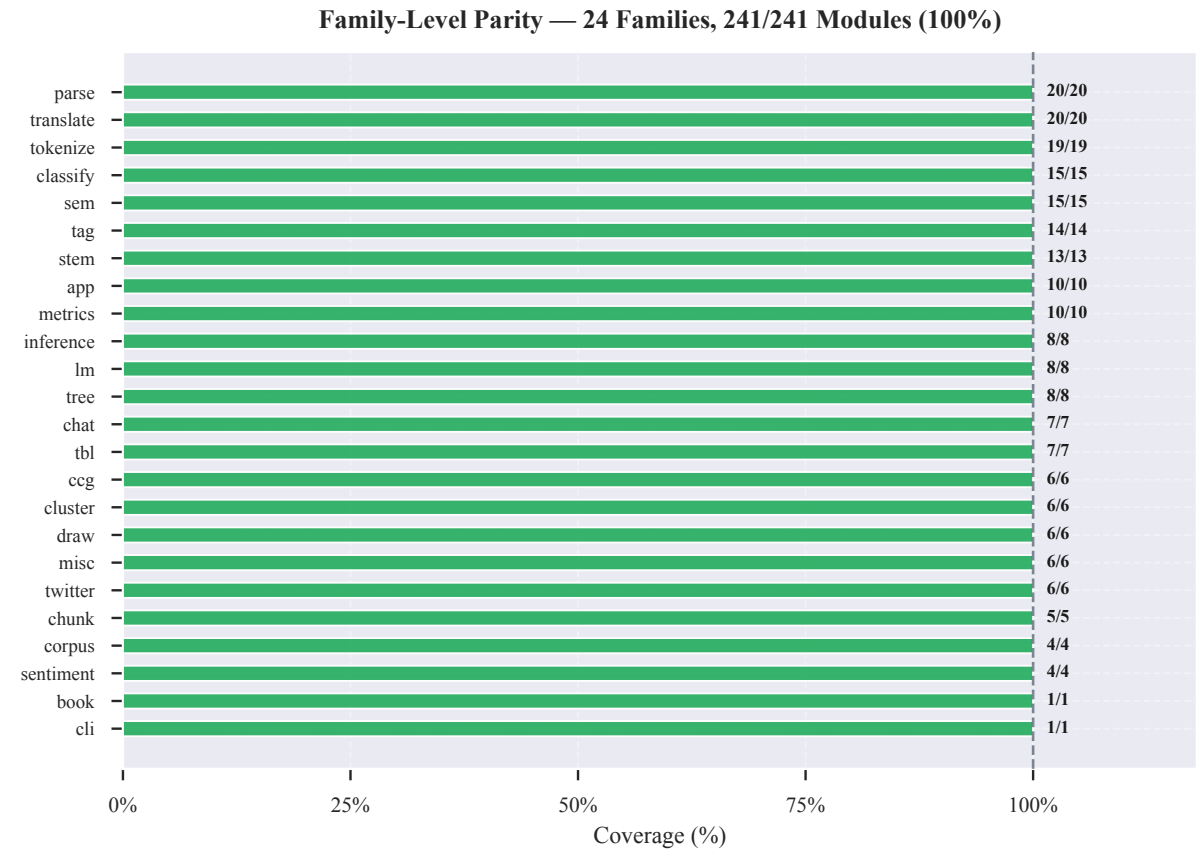


Figure 6: Per-family coverage (all families reach the NLTK API index count; "shim vs. real port" is the split within each filled bar). Green = real port, amber = shim that throws with a JS hint. No family is missing.

Listing 3: Example: CCG chart parse "I sleep" ⇒ S (`examples/ccg_quickstart.ts`).

```
import { fromString } from "bun_nltk/src/ccg_lexicon.ts";
import { CCGChartParser, DefaultRuleSet } from "bun_nltk/src/ccg_chart.ts";

const lex = fromString(‘
:- S, NP, N
NP :: NP
N  :: N
I     => NP
sleep => S\NP
‘);
const parser = new CCGChartParser(lex, DefaultRuleSet);
const parses = parser.parse(["I", "sleep"]);
console.log(parses.length, parses[0].lhs().toString());
// => 1  S
```

Listing 4: Example: FOL resolution proves `mortal(socrates)` (`examples/inference_resolution.ts`).

```
import { LogicParser } from "bun_nltk/src/sem_logic.ts";
import { ResolutionProverCommand } from "bun_nltk/src/inference_resolution.ts";

const lp = new LogicParser();
const assms = [lp.parse("all x.(man(x) -> mortal(x))"), lp.parse("man(socrates)")];
const goal  = lp.parse("mortal(socrates)");
const cmd = new ResolutionProverCommand(goal, assms);
console.log(cmd.prove()); // => true
console.log(cmd.proof());
```